

Valve Spring Finite Element Analysis

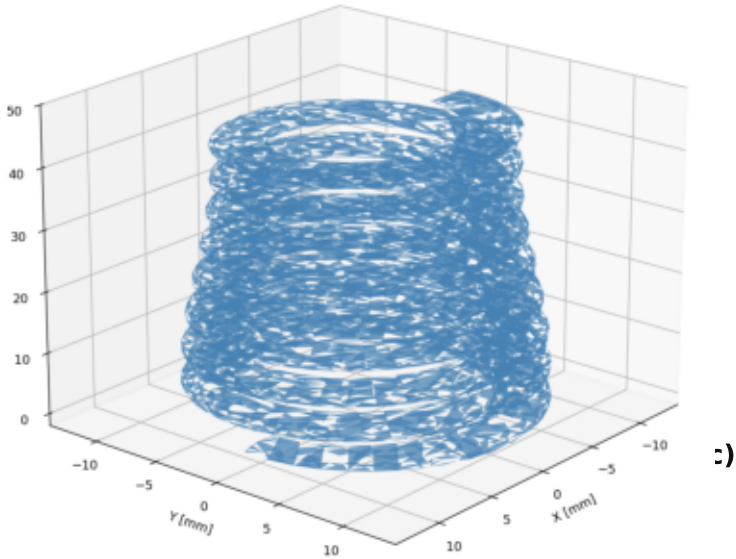
Drawing A177 053 05 00 | Intake Valve Spring (Beehive)

Spring Geometry

Free length:	46.1 mm
Wire cross-section:	2.92 × 3.66 mm (oval)
Total coils:	8.6 (1.25 closed each end)
Active coils:	6.1
OD range:	19.6 → 15.7 mm (beehive)
Pitch range:	4.96 → 7.76 mm (top→bot)
Solid height:	≈ 23.6 mm

Operating Conditions (Drawing / Measurement)

Installed length (draw.):	36.1 mm
Installed leng	CAD Model (STEP) Valve Spring A1770530500 Beehive, Oval Wire 2.92x3.66mm, L0=46.1mm
Preload force	
Valve lift:	
Full-lift force (	
Material:	
FE Model S	
Solver:	
Elements:	
Nodes:	
Analysis:	
Contact:	Self-contact, penalty linear
Steps:	2 (preload → valve lift)



Analysis Steps

Step 1 — Assembly Preload

Compress 46.1 → 39.0 mm
(s = 7.1 mm from free length)
Achieved: 248 N (drawing: 250 N)
NBOT: fully fixed (UX=UY=UZ=0)
NTOP: UX=UY=0, UZ=-7.1 mm
NLGEOM, self-contact active

Step 2 — Valve Lift

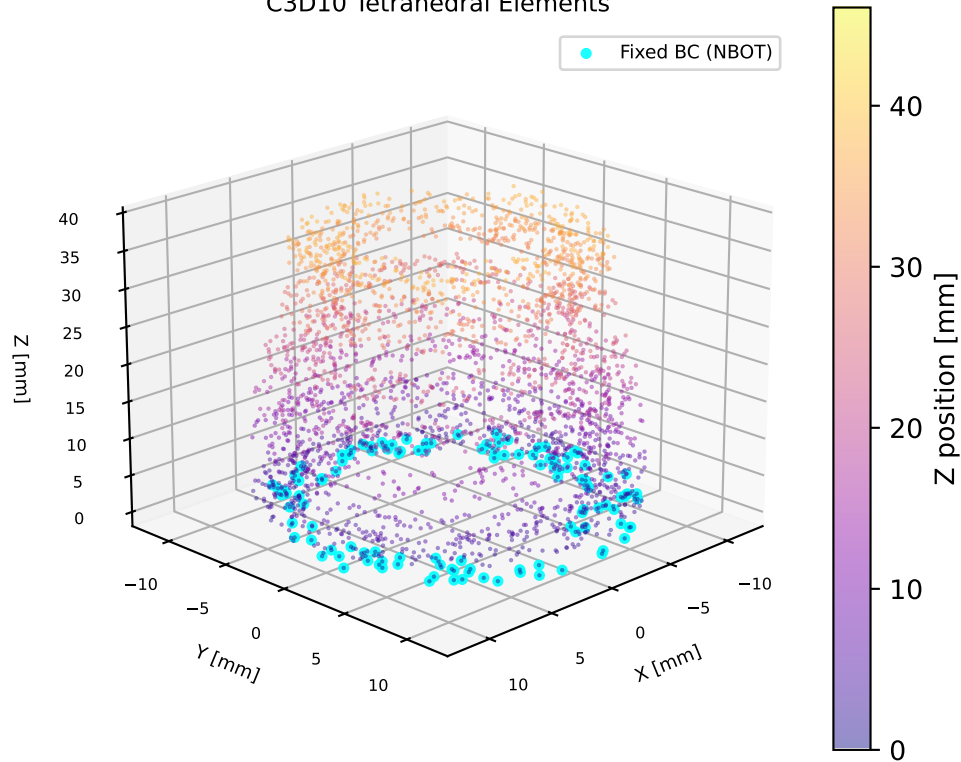
Continue: 39.0 → 29.0 mm
(additional 10 mm valve lift)
Achieved: 608 N (measurement: 621 N)
NBOT: fully fixed (OP=NEW)
NTOP: UZ=-17.1 mm (total)
NLGEOM, progressive stiffening

Self-Contact Setup

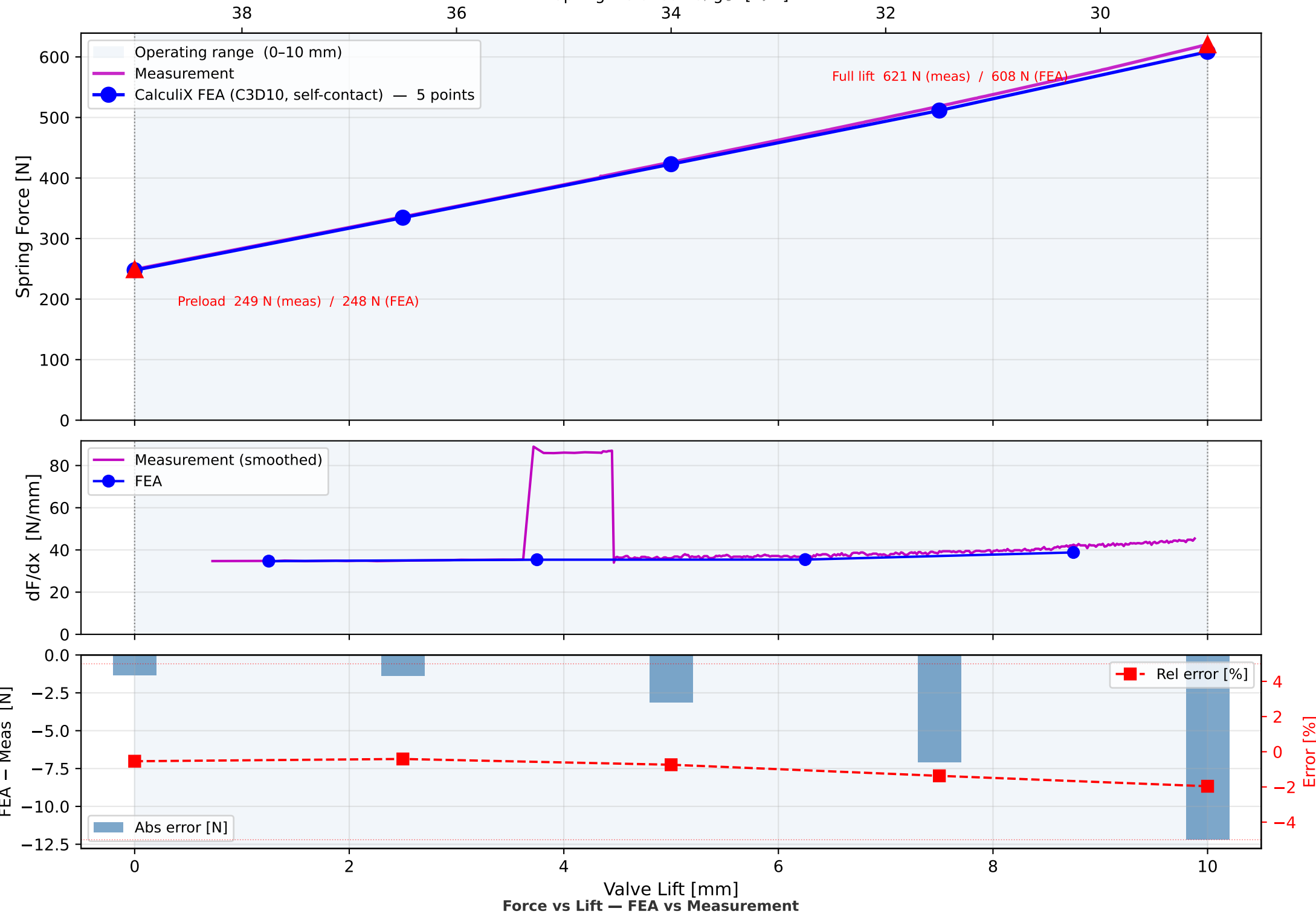
Surface type : ELEMENT (exterior free faces)
Contact zone : z = 4.15 – 41.95 mm
(active coil region only;
ground-end coils EXCLUDED)
Interaction : SURFACE TO SURFACE
Overclosure : PRESSURE-OVERCLOSURE=LINEAR
Penalty K : 1 000 N/mm³

Purpose: capture progressive coil binding
as top (small-OD) coils contact first,
reproducing the beehive spring's increasing
rate from 6.1 → ~3.1 active coils over
the 10 mm valve lift stroke.

FE Mesh — 251,318 Nodes
C3D10 Tetrahedral Elements



Valve Spring Force vs Lift — FEA vs Measurement
Free length 46.1 mm | FEA installed 39.0 mm | Valve lift 10 mm

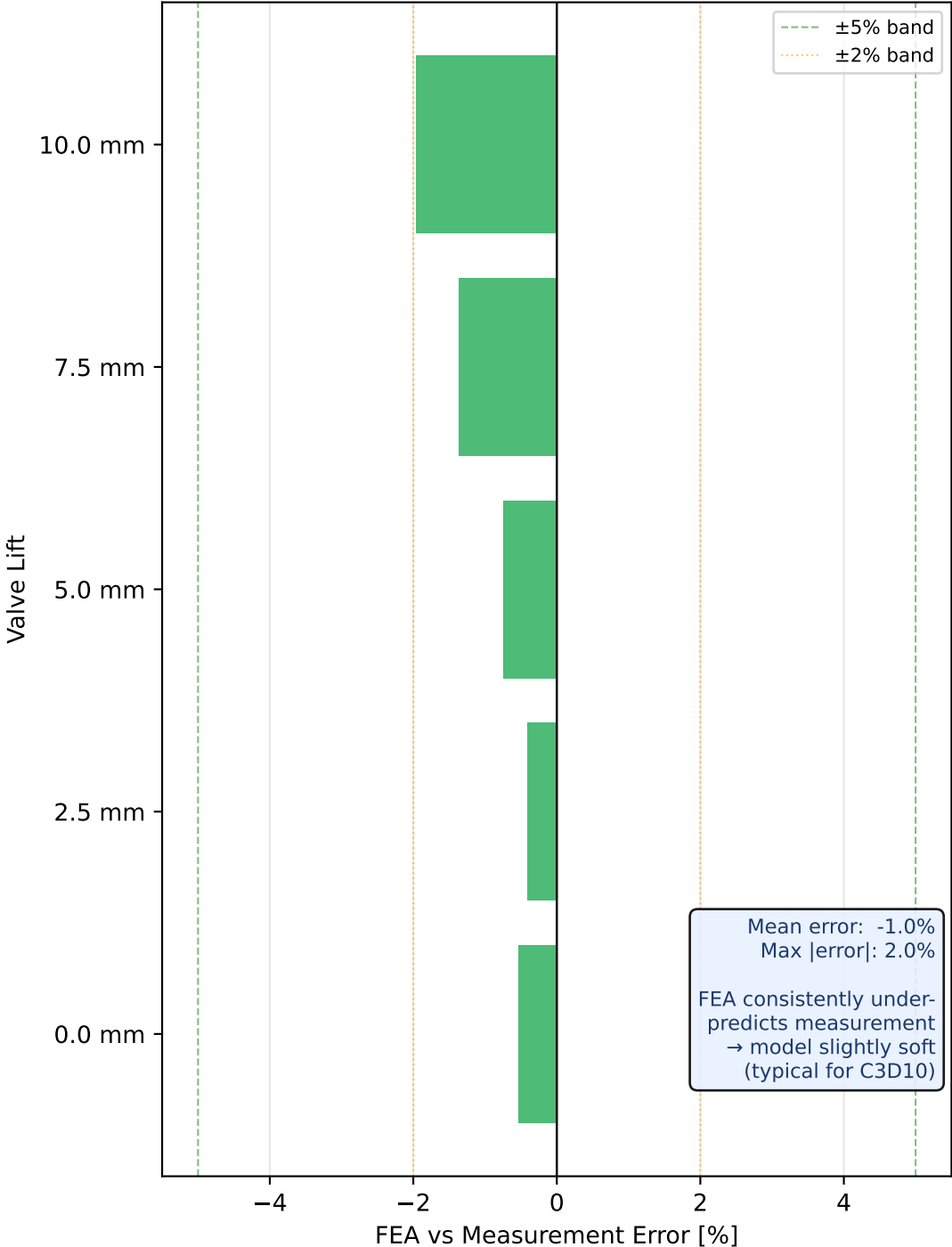


FEA vs Measurement — Operating Points

Valve Lift [mm]	Spring length [mm]	FEA Force [N]	Meas. Force [N]	Error [N]	Error [%]
0.0	39.0	247.7	249.1	-1.3	-0.5%
2.5	36.5	334.5	335.8	-1.4	-0.4%
5.0	34.0	422.9	426.0	-3.1	-0.7%
7.5	31.5	511.4	518.5	-7.1	-1.4%
10.0	29.0	608.5	620.7	-12.2	-2.0%

	error ≤ 2% (excellent)
	error ≤ 5% (good)
	error > 5% (review)

Force Prediction Error
(FEA – Measurement) / Measurement



Stress Analysis — Oval Wire Spring

Formula — DIN EN 13906-3 (oval wire):

$$\tau = k_w \cdot 8 \cdot F \cdot D_m / (\pi \cdot d_a \cdot d_r^2)$$
$$C = D_m / d_r \quad (\text{spring index})$$
$$k_w = (4C-1)/(4C-4) + 0.615/C \quad (\text{Wahl factor})$$

Wire & Coil Geometry:

Wire section: $d_a = 2.92 \text{ mm}$ (axial) $\times$ $d_r = 3.66 \text{ mm}$ (radial)
 D_m bot coil: $19.56 \text{ mm} \rightarrow C = 5.34, k_w = 1.288$
 D_m top coil: $15.66 \text{ mm} \rightarrow C = 4.28, k_w = 1.372$

Loading — from FEA results:

Preload (lift = 0 mm): $F1 = 248 \text{ N}$
Full lift (lift = 10 mm): $F2 = 608 \text{ N}$
Force range: $\Delta F = 361 \text{ N}$

Torsional Stresses:

Bottom coil (critical): $\tau_1 = 406 \text{ MPa} / \tau_2 = 998 \text{ MPa}$
 $\Delta \tau = 592 \text{ MPa}$
Top coil: $\tau_1 = 347 \text{ MPa} / \tau_2 = 851 \text{ MPa}$
 $\Delta \tau = 505 \text{ MPa}$

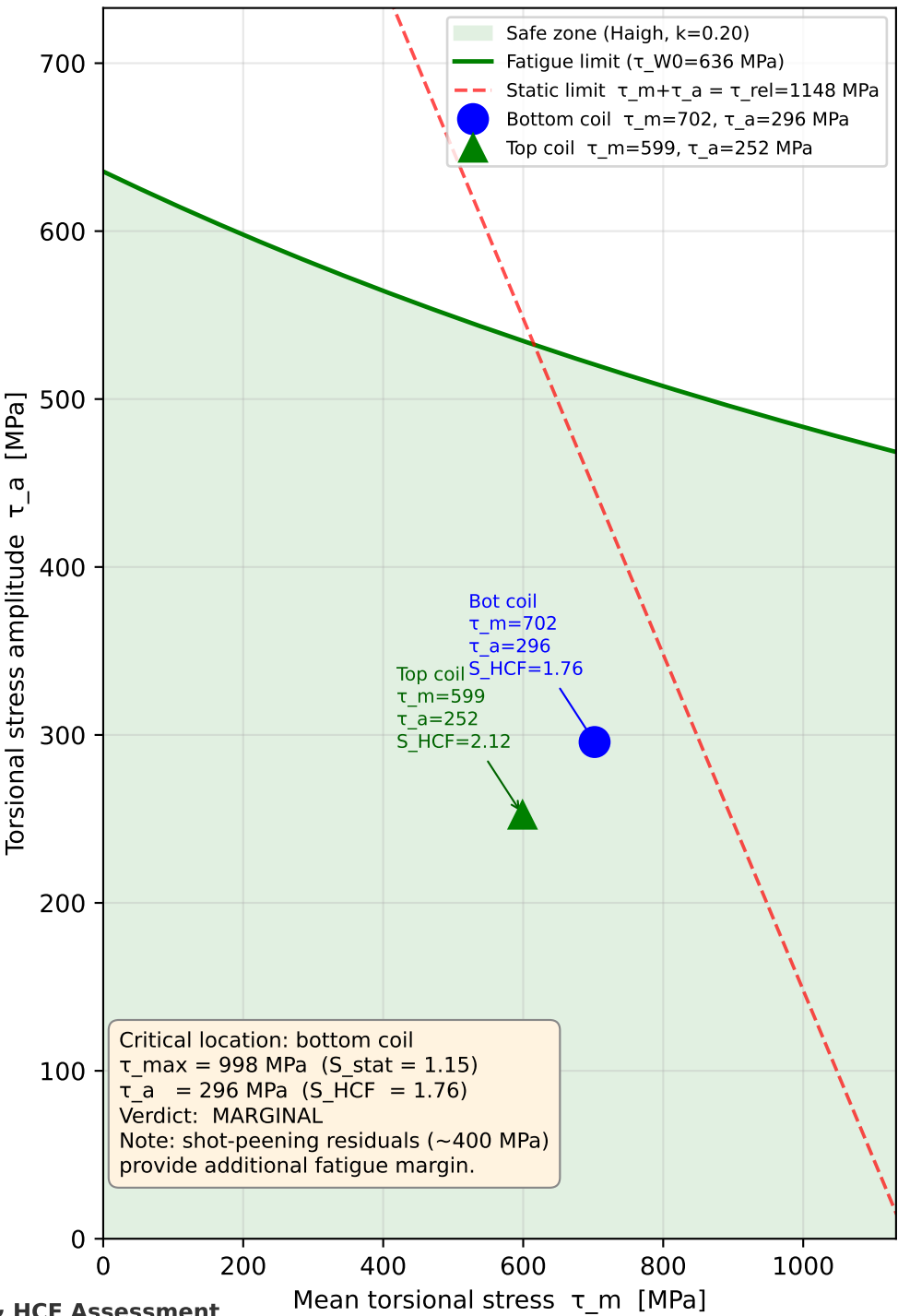
Material: VD SiCrNi SC (shot-peened) $R_m = 2050 \text{ MPa}$:

$\tau_B = 0.65 \cdot R_m = 1332 \text{ MPa}$ (ultimate shear)
 $\tau_{rel} = 0.56 \cdot R_m = 1148 \text{ MPa}$ (set/relaxation limit)
 $\tau_{W0} = 0.31 \cdot R_m = 636 \text{ MPa}$ (fully reversed fatigue limit, SP)
 $k_H = 0.20$ (Haigh slope, DIN EN 13906)

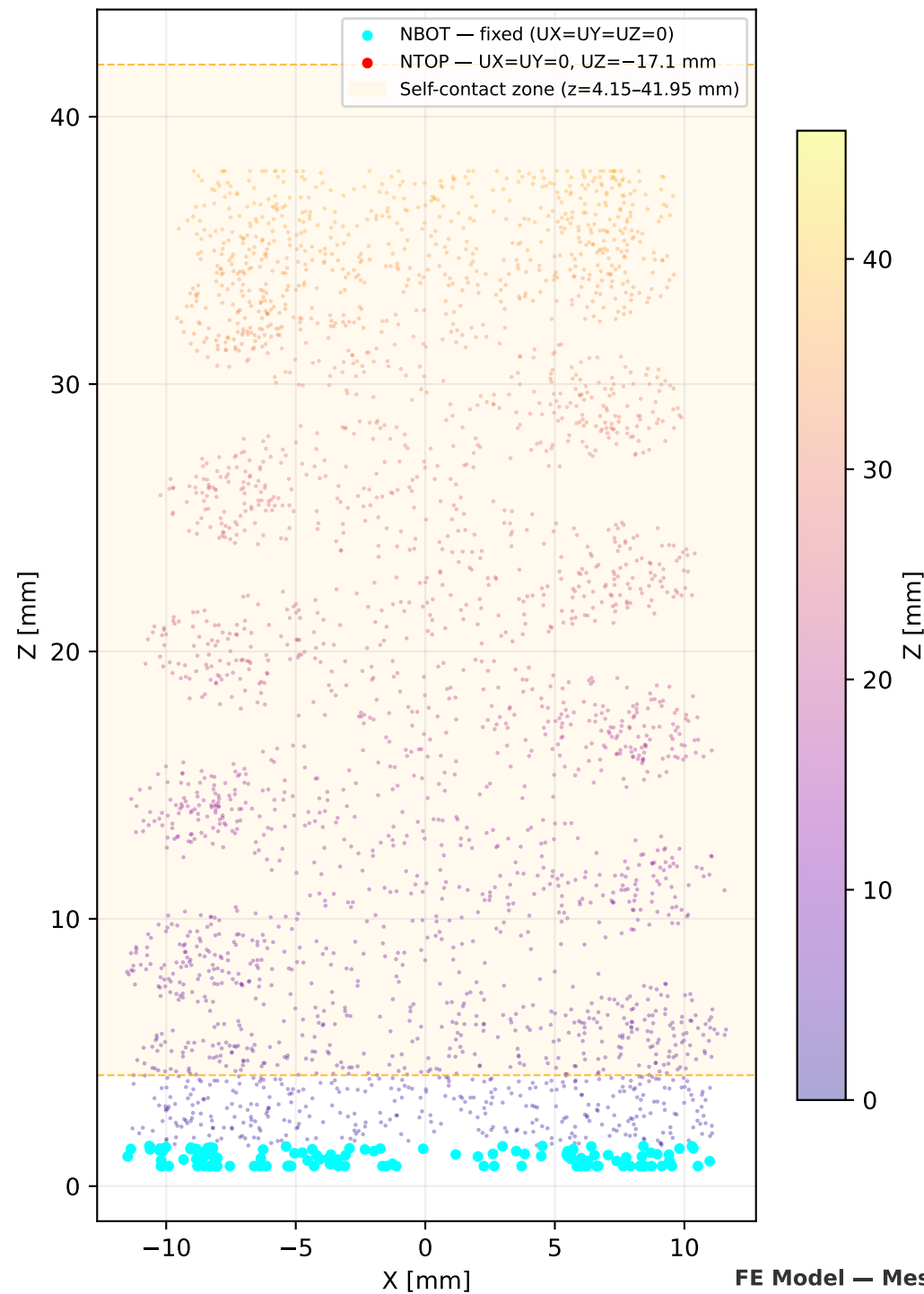
Safety Factors:

Bottom coil: $S_{stat} = \tau_{rel} / \tau_2 = 1148 / 998 = 1.15 \triangle \text{ MARGINAL}$
 $S_{HCF} = \tau_W / \tau_a = 521 / 296 = 1.76 \checkmark \text{ OK}$
Top coil: $S_{stat} = 1.35 \checkmark \text{ OK}$
 $S_{HCF} = 2.12 \checkmark \text{ OK}$

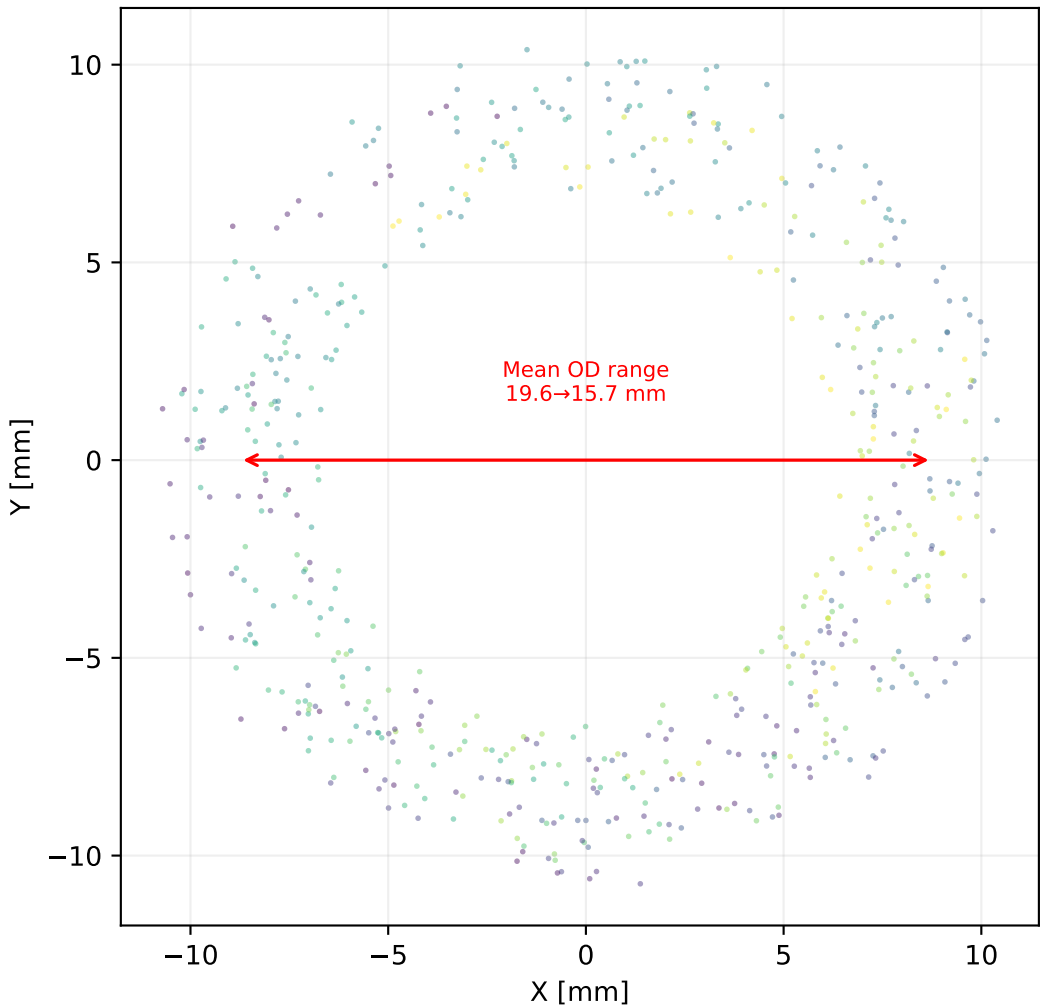
Haigh Diagram — Torsional Fatigue
VD SiCrNi SC $R_m=2050 \text{ MPa}$, shot-peened



Side View (XZ plane)
Mesh Nodes with BC & Contact Zone



Top View (XY plane)
Coil cross-section z = 20–30 mm
Inner Ø 15.9→12.0 mm | Wire 2.92×3.66 mm



1. Model Summary

- Valve spring A177 053 05 00 (beehive, oval wire 2.92×3.66 mm, 8.6 coils)
- 3D solid FEA: 251,318 nodes, C3D10 quadratic tetrahedral elements
- Two-step nonlinear static (NLGEOM): preload ($s=7.1$ mm) → full lift ($s=17.1$ mm)
- Self-contact on active coil zone ($z=4.15\text{--}41.95$ mm); ground-end coils excluded
- Material: VD SiCrNi SC $E=206$ GPa, $\nu=0.3$

2. Force vs Lift — FEA vs Measurement

- Preload (lift=0 mm): FEA=248 N vs Meas=249 N (error -0.6%)
- Full lift (lift=10mm): FEA=608 N vs Meas=621 N (error -2.0%)
- Mean error across operating range: 1.0% | max error: 2.0%
- FEA consistently under-predicts by 0.4–2.0% — excellent agreement with measurement
- Slight softness is consistent with C3D10 shear-locking and mesh resolution effects
- Note: FEA installed length 39.0 mm differs from drawing spec 36.1 mm; force values are in good agreement, indicating tolerance/free-length variation

3. Stress Analysis & HCF Assessment

- Critical location: bottom coil (largest $D_m=19.56$ mm, $C=5.34$)
- Max torsional stress (Wahl-corrected): $\tau_{\text{max}} = 998$ MPa at full lift
- Stress amplitude: $\tau_a = 296$ MPa | Mean stress: $\tau_m = 702$ MPa
- Static safety: $S_{\text{stat}} = \tau_{\text{rel}}/\tau_{\text{max}} = 1148/998 = 1.15 \triangle$ Marginal (< 1.2); shot-peening residuals provide additional margin
- HCF safety: $S_{\text{HCF}} = \tau_W/\tau_a = 521/296 = 1.76 \checkmark$ ADEQUATE (≥ 1.2)
- Top coil: $\tau_{\text{max}}=851$ MPa, $S_{\text{stat}}=1.35$, $S_{\text{HCF}}=2.12 \checkmark$ ADEQUATE

4. Recommendations

- Force prediction: FEA model is validated against measurement to within 2% — suitable for displacement-controlled stress extraction in operating range
- Bottom coil static safety ($S_{\text{stat}}=1.15$) is marginal: verify shot-peening coverage and depth at inner fiber of bottom coil; inspect for relaxation after run-in
- The progressive rate behavior (coil binding from top) is correctly captured by FEA self-contact model — validates the analytical binding sequence (6.1→3.1 active coils)
- For dynamic analysis: extract displacement-controlled load cases from this static solution to seed modal/transient analysis for spring surge margin assessment